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Structure of Atom

NEET Chemistry · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	92 questions
Answer key	at the end

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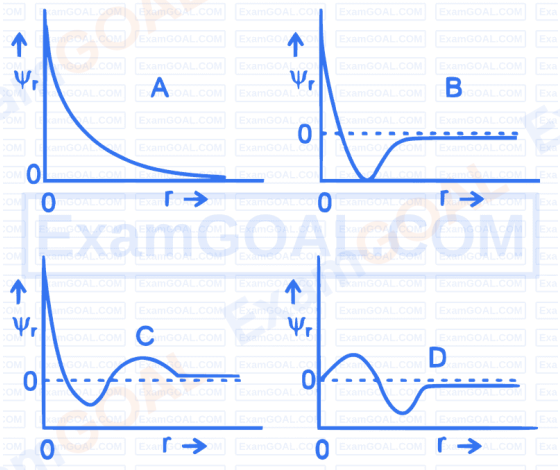
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MCQ (Single Correct Answer) 92 questions

Q1. RE-NEET 2026

Open solution

Consider the following schematic plots of orbital wavefunction (ψ_r) against distance (r) from the nucleus.



The figure representing two radial nodes in the orbital is

(A) D (B) A (C) B (D) C

Q2. NEET 2026

Open solution

Match List I with List II :

	List I (Quantum Numbers)			List II (Orbital)
	'n'	'l'		
A.	2	1	I.	3d
B.	4	0	II.	2p
C.	5	3	III.	4s
D.	3	2	IV.	5f

Choose the correct answer from the options given below.

(A) A-IV, B-II, C-III, D-I (B) A-II, B-III, C-I, D-IV

(C) A-II, B-III, C-IV, D-I (D) A-I, B-II, C-III, D-IV

Q3. NEET 2026

Open solution

A bulb is rated at 150 watt, converting 8% energy into light. If energy of one photon is 4.42×10^{-19} J, how many photons are emitted by the bulb per second?

(A) 2.71×10^{19} (B) 4.06×10^{19} (C) 27.2×10^{19} (D) 1.35×10^{19}

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Q4. NEET 2025

The ratio of the wavelengths of the light absorbed by a Hydrogen atom when it undergoes $n = 2 \rightarrow n = 3$ and $n = 4 \rightarrow \mathrm{n} = 6$ transitions, respectively, is

(A) $\frac{1}{9}$ (B) $\frac{1}{4}$ (C) $\frac{1}{36}$ (D) $\frac{1}{16}$

Open solution

Q5. NEET 2025

Energy and radius of first Bohr orbit of $He^{+}$and Li^{2+} are [Given $\mathrm{R_H} = 2.18 \times 10^{-18}\ \mathrm{J}, \mathrm{a_0} = 52.9\mathrm{pm}$]

(A) $\mathrm{E_n}\left(\mathrm{Li}^{2+}\right) = -19.62 \times 10^{-16}\ \mathrm{J};$
$\mathrm{r_n}\left(\mathrm{Li}^{2+}\right) = 17.6\mathrm{pm}$
$\mathrm{E_n}\left(\mathrm{He}^{+}\right) = -8.72 \times 10^{-16}\ \mathrm{J};$
$\mathrm{r_n}\left(\mathrm{He}^{+}\right) = 26.4\mathrm{pm}$

(B) $\mathrm{E_n}\left(\mathrm{Li}^{2+}\right) = -8.72 \times 10^{-16}\ \mathrm{J};$
$\mathrm{r_n}\left(\mathrm{Li}^{2+}\right) = 17.6\mathrm{pm}$
$\mathrm{E_n}\left(\mathrm{He}^{+}\right) = -19.62 \times 10^{-16}\ \mathrm{J};$
$\mathrm{r_n}\left(\mathrm{He}^{+}\right) = 17.6\mathrm{pm}$

(C) $\mathrm{E_n}\left(\mathrm{Li}^{2+}\right) = -19.62 \times 10^{-18}\ \mathrm{J};$
$\mathrm{r_n}\left(\mathrm{Li}^{2+}\right) = 17.6\mathrm{pm}$
$\mathrm{E_n}\left(\mathrm{He}^{+}\right) = -8.72 \times 10^{-18}\ \mathrm{J};$
$\mathrm{r_n}\left(\mathrm{He}^{+}\right) = 26.4\mathrm{pm}$

(D) $\mathrm{E_n}\left(\mathrm{Li}^{2+}\right) = -8.72 \times 10^{-18}\ \mathrm{J};$
$\mathrm{r_n}\left(\mathrm{Li}^{2+}\right) = 26.4\mathrm{pm}$
$\mathrm{E_n}\left(\mathrm{He}^{+}\right) = -19.62 \times 10^{-18}\ \mathrm{J};$
$\mathrm{r_n}\left(\mathrm{He}^{+}\right) = 17.6\mathrm{pm}$

Open solution

Q6. NEET 2024 (Re-Examination)

The quantum numbers of four electrons are given below :

I. $n = 4; l = 2; m_l = -2; s = -\frac{1}{2}$

II. $n = 3; l = 2; m_l = 1; s = +\frac{1}{2}$

III. $n = 4; l = 1; m_l = 0; s = +\frac{1}{2}$

IV. $n = 3; l = 1; m_l = -1; s = +\frac{1}{2}$

The correct decreasing order of energy of these electrons is

(A) IV > II > III > I (B) I > III > II > IV

(C) III > I > II > IV (D) I > II > III > IV

Open solution

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Q7. NEET 2024 (Re-Examination)

Open solution

Given below are two statements:

Statement I: The Balmer spectral line for H atom with lowest energy is located at $\frac{5}{36}R_H\ \mathrm{cm}^{-1}$.

($\mathrm{R_H}$ = Rydberg constant)

Statement II: When the temperature of blackbody increases, the maxima of the curve (intensity and wavelength) shifts to shorter wavelength.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q8. NEET 2024

Open solution

The energy of an electron in the ground state $(\mathrm{n}=1)$ for He^{+} ion is $-\mathrm{x}\ \mathrm{J}$, then that for an electron in $(\mathrm{n}=2)$ state for Be^{3+} ion in J is

(A) $-x$ (B) $-\frac{x}{9}$ (C) $-4x$ (D) $-\frac{4}{9}x$

Q9. NEET 2024

Open solution

Match List I with List II.

	List I (Quantum Number)		List II (Information provided)
A.	$\mathrm{m_l}$	I.	Shape of orbital
B.	$\mathrm{m_s}$	II.	Size of orbital
C.	l	III.	Orientation of orbital
D.	n	IV.	Orientation of spin of electron

Choose the correct answer from the options given below :

(A) A-I, B-III, C-II, D-IV (B) A-III, B-IV, C-I, D-II

(C) A-III, B-IV, C-II, D-I (D) A-II, B-I, C-IV, D-III

Q10. NEET 2023 Manipur

Open solution

Incorrect set of quantum numbers from the following is :

(A) $\mathrm{n}=4, \ell=3,\ \mathrm{m}_\ell=-3,-2,-1,0,+1,+2,+3,\ \mathrm{m_s}=-1/2$

(B) $\mathrm{n}=5, \ell=2,\ \mathrm{m}_\ell=-2,-1,+1,+2,\ \mathrm{m_s}=+1/2$

(C) $\mathrm{n}=4, \ell=2,\ \mathrm{m}_\ell=-2,-1,0,+1,+2, \mathrm{m_s}=-1/2$

(D) $\mathrm{n}=5, \ell=3,\ \mathrm{m}_\ell=-3,-2,-1,0,+1,+2,+3, \mathrm{m_s}=+1/2$

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Q11. NEET 2023 Manipur

Open solution

Given below are two statements:

Statement I : The value of wave function, ψ depends upon the coordinates of the electron in the atom.

Statement II : The probability of finding an electron at a point within an atom is proportional to the orbital wave function.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is true but Statement II is false.

(B) Statement I is false but Statement II is true.

(C) Both Statement I and Statement II are true.

(D) Both Statement I and Statement II are false.

Q12. NEET 2023

Open solution

Select the correct Statements from the following :

A. Atoms of all elements are composed of two fundamental particles.

B. The mass of the electron is 9.10939×10^{-31} kg.

C. All the isotopes of a given elements show same chemical properties.

D. Protons and electrons are collectively known as nucleons.

E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

Choose the correct answer from the options given below.

(A) C, D and E only (B) A and E only (C) B, C and E only (D) A, B and C only

Q13. NEET 2023

Open solution

The relation between $\mathrm{n_m}$, ($\mathrm{n_m}$ = the number of permissible values of magnetic quantum number (m)) for a given value of azimuthal quantum number (l), is

(A) $l = 2\mathrm{n_m} + 1$ (B) $\mathrm{n_m} = 2l^2 + 1$

(C) $\mathrm{n_m} = l + 2$ (D) $l = \frac{\mathrm{n_m} - 1}{2}$

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Q14. NEET 2022 Phase 2

Open solution

Match List-I with List-II :

	List-I (quantum number)		List-II (orbital)
(a)	n = 2, l = 1	(i)	2s
(b)	n = 3, l = 2	(ii)	3s
(c)	n = 3, l = 0	(iii)	2p
(d)	n = 2, l = 0	(iv)	3d

Choose the correct answer from the options given below :

(A) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

(B) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

(C) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(D) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Q15. NEET 2022 Phase 2

Open solution

When electromagnetic radiation of wavelength 300 nm falls on the surface of a metal, electrons are emitted with the kinetic energy of 1.68×10^5 J mol^{-1}. What is the minimum energy needed to remove an electron from the metal?

($h = 6.626 \times 10^{-34}$ Js, $c = 3 \times 10^8$ ms^{-1}, $N_A = 6.022 \times 10^{23}$ mol^{-1})

(A) 2.31×10^5 J mol^{-1}

(B) 2.31×10^6 J mol^{-1}

(C) 3.84×10^4 J mol^{-1}

(D) 3.84×10^{-19} J mol^{-1}

Q16. NEET 2022 Phase 1

Open solution

Identify the incorrect statement from the following.

(A) All the five 5d orbitals are different in size when compared to the respective 4d orbitals.

(B) All the five 4d orbitals have shapes similar to the respective 3d orbitals.

(C) In an atom, all the five 3d orbitals are equal in energy in free state.

(D) The shapes of d_{xy}, d_{yz} and d_{zx} orbitals are similar to each other; and $d_{x^2-y^2}$ and d_{z^2} are similar to each other.

Q17. NEET 2022 Phase 1

Open solution

If radius of second Bohr orbit of the He^+ ion is 105.8 pm, what is the radius of third Bohr orbit of Li^{2+} ion?

(A) 158.7 pm **(B)** 15.87 pm **(C)** 1.587 pm **(D)** 158.7 $\overset{o}{A}$

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Q18. NEET 2021

A particular station of All India Radio, New Delhi broadcasts on a frequency of 1,368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is : [speed of light c = 3.0×10^8 ms^{-1}]

(A) 21.92 cm (B) 219.3 m (C) 219.2 m (D) 2192 m

Open solution

Q19. NEET 2021

From the following pairs of ions which one is not an iso-electronic pair?

(A) Fe^{2+}, Mn^{2+} (B) O^{2-}, F^- (C) Na^+, Mg^{2+} (D) Mn^{2+}, Fe^{3+}

Open solution

Q20. NEET 2020 Phase 2

The number of angular nodes and radial nodes in $3s$ orbital are

(A) 0 and 1, respectively (B) 0 and 2, respectively
(C) 1 and 0, respectively (D) 3 and 0, respectively

Open solution

Q21. NEET 2019 (Odisha)

Orbital having 3 angular nodes and 3 total nodes is

(A) 6d (B) 5p (C) 3d (D) 4f

Open solution

Q22. NEET 2019 (Odisha)

In hydrogen atom, the de Broglie wavelength of an electron in the second Bohr orbit is
[Given that Bohr radius, $a_0 = 52.9$ pm]

(A) 105.8 pm (B) 211.6 pm (C) 211.6π pm (D) 52.9π pm

Open solution

Q23. NEET 2019

4d, 5p, 5f and 6p orbitals are arranged in the order of decreasing energy. The correct option is :

(A) 6p > 5f > 4d > 5p (B) 5f > 6p > 4d > 5p
(C) 5f > 6p > 5p > 4d (D) 6p > 5f > 5p > 4d

Open solution

Q24. NEET 2019

Which of the following series of transitions in the spectrum of hydrogen atom falls in visible region?

(A) Paschen series (B) Brackett series (C) Lyman series (D) Balmer series

Open solution

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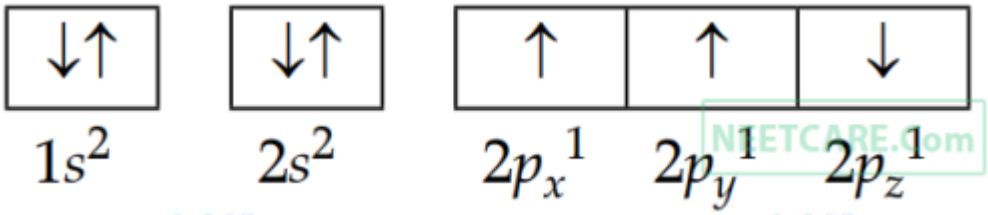
Q25. NEET 2018

Which one is a wrong statement?

(A) Total orbital angular momentum of electron in s-orbital is equal to zero.

(B) An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers

(C) The electronic configuration of N atom is



(D) The value of m for dz^2 is zero.

Open solution

Q26. NEET 2017

Which one is the wrong statement ?

(A) The uncertainty principle is

$\Delta E \times \Delta t \geq \frac{h}{4\pi}$

(B) Half filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.

(C) The energy of 2s-orbital is less than the energy of 2p-orbital in case of hydrogen like atoms.

(D) de-Broglie's wavelength is given by

$\lambda = \frac{h}{mv}$, where m = mass of the particle,

v = group velocity of the particle.

Open solution

Q27. NEET 2016 Phase 2

How many electrons can fit in the orbital for which n = 3 and l = 1?

(A) 2 (B) 6 (C) 10 (D) 14

Open solution

Q28. NEET 2016 Phase 2

Which of the following pairs of d-orbitals will have electron density along the axes ?

(A) d_{z^2}, d_{xz} (B) d_{xz}, d_{yz}

(C) d_{z^2}, $d_{x^2-y^2}$ (D) d_{xy}, $d_{x^2-y^2}$

Open solution

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Q29. NEET 2016 Phase 1

Two electrons occupying the same orbital are distinguished by

(A) azimuthal quantum number
(B) spin quantum number
(C) principal quantum number
(D) magnetic quantum number

Open solution

Q30. AIPMT 2015

Which is the correct order of increasing energy of the listed orbitals in the atom of titanium ? (At. no. Z = 22)

(A) 4s 3s 3p 3d
(B) 3s 3p 3d 4s
(C) 3s 3p 4s 3d
(D) 3s 4s 3p 3d

Open solution

Q31. AIPMT 2015 Cancelled Paper

The number of d-electrons in Fe^{2+} (Z = 26) is not equal to the number of electrons in which one of the following ?

(A) d-electrons in Fe (Z = 26)
(B) p-electrons in Ne (Z = 10)
(C) s-electrons in Mg (Z = 12)
(D) p-electrons in Cl(Z = 17)

Open solution

Q32. AIPMT 2015 Cancelled Paper

The angular momentum of electron in 'd' orbital is equal to

(A) $2\sqrt{3}h$
(B) 0 $\hbar$
(C) $\sqrt{6}\,\hbar$
(D) $\sqrt{2}\,\hbar$

Open solution

Q33. AIPMT 2014

What is the maximum number of orbitals that can be identified with the following quantum numbers ?
n = 3, l = 1, m_1 = 0

(A) 1
(B) 2
(C) 3
(D) 4

Open solution

Q34. AIPMT 2014

Calculate the energy in joule corresponding to light of wavelength 45 nm. (Planck,s constant, h = 6.63×10^{-34} J s, speed of light, c = 3×10^{8} m s^{-1})

(A) 6.67×10^{15}
(B) 6.67×10^{11}
(C) 4.42×10^{-15}
(D) 4.42×10^{-18}

Open solution

Q35. AIPMT 2014

Be^{2+} is isoelectronic with which of the following ions ?

(A) H^+
(B) Li^+
(C) Na^+
(D) Mg^{2+}

Open solution

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Q36. NEET 2013 (Karnataka)

The outer electronic configuration of Gd (At. No. 64) is

(A) $4f^5\ 5d^4\ 6s^1$ (B) $4f^7\ 5d^1\ 6s^2$ (C) $4f^3\ 5d^5\ 6s^2$ (D) $4f^4\ 5d^5\ 6s^1$

Open solution

Q37. NEET 2013 (Karnataka)

According to law of photochemical equivalence the energy absorbed (in ergs/mole) is given as (h = 6.62×10^{-27} ergs, c = 3×10^{10} cm s^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1})

(A) $\frac{1.196\times10^{8}}{\lambda}$ (B) $\frac{2.859\times10^{5}}{\lambda}$

(C) $\frac{2.859\times10^{16}}{\lambda}$ (D) $\frac{1.196\times10^{16}}{\lambda}$

Open solution

Q38. NEET 2013

What is the maximum numbers of electrons that can be associated with the following set of quantum numbers ?

n = 3, l = 1 and m = -1

(A) 4 (B) 2 (C) 10 (D) 6

Open solution

Q39. NEET 2013

Based on equation E = -2.178×10^{-18} J $\left(\frac{Z^2}{n^2}\right)$, certain conclusions are written. Which of them is not correct ?

(A) Equation can be used to calculate the change in energy when the electron changes orbit.

(B) For n = 1, the electron has a more negative energy than it does for n = 6 which means that the electron is more loosely bound in the smallest allowed orbit.

(C) The negative sign in equation simply means that the energy of electron bound to the nucleus is lower than it would be if the electrons were at the infinite distance from the nucleus.

(D) Larger the value of n, the larger is the orbit radius.

Open solution

Q40. NEET 2013

The value of Planck's constant is 6.63×10^{-34} J s. The speed of light is 3×10^{17} nm s^{-1}. Which value is closest to the wavelength in nanometer of a quantum of light with frequency of 6×10^{15} s^{-1} ?

(A) 50 (B) 75 (C) 10 (D) 25

Open solution

Q41. AIPMT 2012 Mains

The orbital angular momentum of a p-electron is given as

(A) $\frac{h}{\sqrt{2\pi}}$ (B) $\sqrt{3}\frac{h}{2\pi}$

(C) $\sqrt{\frac{3}{2}}\frac{h}{\pi}$ (D) $\sqrt{6}\frac{h}{2\pi}$

Open solution

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Q42. AIPMT 2012 Prelims

Maximum number of electrons in a subshell with l = 3 and n = 4 is

(A) 14 (B) 16 (C) 10 (D) 12

Open solution

Q43. AIPMT 2012 Prelims

The correct set of four quantum numbers for the valence electron of rubidium atom (Z = 37) is

(A) 5, 1, 1, + 1/2 (B) 6, 0, 0, +1/2 (C) 5, 0, 0, +1/2 (D) 5, 1, 0, +1/2

Open solution

Q44. AIPMT 2011 Mains

According to the Bohr theory, which of the following transitions in the hydrogen atom will give rise to the least energetic photon ?

(A) n = 6 to n = 1 (B) n = 5 to n = 4 (C) n = 6 to n = 5 (D) n = 5 to n = 3

Open solution

Q45. AIPMT 2011 Prelims

The total number of atomic orbitals in fourth energy level of an atom is

(A) 8 (B) 16 (C) 32 (D) 4

Open solution

Q46. AIPMT 2011 Prelims

The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relation between their wavelengths i.e., λ_1 and λ_2 will be

(A) $\lambda_1 = \lambda_2$ (B) $\lambda_1 = 2\lambda_2$

(C) $\lambda_1 = 4\lambda_2$ (D) $\lambda_1 = \frac{1}{2}\lambda_2$

Open solution

Q47. AIPMT 2011 Prelims

If n = 6, the correct sequence for filling of electrons will be

(A) $ns \rightarrow (n-2)f \rightarrow (n-1)d \rightarrow np$ (B) $ns \rightarrow (n-1)d \rightarrow (n-2)f \rightarrow np$

(C) $ns \rightarrow (n-2)f \rightarrow np \rightarrow (n-1)d$ (D) $ns \rightarrow np\,(n-1)d \rightarrow (n-2)f$

Open solution

Q48. AIPMT 2010 Mains

A 0.66 kg ball is moving with a speed of 100 m/s. The associated wavelength will be

(h = 6.6×10^{-34} J s)

(A) 6.6×10^{-32} m (B) 6.6×10^{-34} m

(C) 1.0×10^{-35} m (D) 1.0×10^{-32} m

Open solution

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Q49. AIPMT 2009

Maximum number of electrons in a subshell of an atom is determined by the following

(A) $2l + 1$ (B) $4l - 2$ (C) $2n^2$ (D) $4l + 2$

Open solution

Q50. AIPMT 2009

Which of the following is not permissible arrangement of electrons in an atom ?

(A) n = 5, l = 3, m = 0, s = +1/2 (B) n = 3, l = 2, m = $-$ 3, s = $-1/2$

(C) n = 3, l = 2, m = $-$ 2, s = $-1/2$ (D) n = 4, l = 0, m = 0, s = $-1/2$

Open solution

Q51. AIPMT 2008

If uncertainty in position and momentum are equal, then uncertainty in velocity is

(A) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$ (B) $\sqrt{\frac{h}{\pi}}$

(C) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ (D) $\sqrt{\frac{h}{2\pi}}$

Open solution

Q52. AIPMT 2008

The measurement of the electron position is associated with an uncertainty in momentum, which is equal to 1×10^{-18} g cm s^{-1}. The uncertainty in electron velocity is (mass of an electron is 9×10^{-28} g)

(A) 1×10^5 cm s^{-1} (B) 1×10^{11} cm s^{-1}

(C) 1×10^9 cm s^{-1} (D) 1×10^6 cm s^{-1}

Open solution

Q53. AIPMT 2007

Consider the following sets of quantum numbers :

	n		l		m		s
(i)	3		0		0		+1/2
(ii)	2		2		1		+1/2
(iii)	4		3		-2		-1/2
(iv)	1		0		-1		-1/2
(v)	3		2		3		+1/2

Open solution

Which of the following sets of quantum number is not possible ?

(A) (i), (ii), (iii) and (iv) (B) (ii), (iv) and (v)

(C) (i) and (iii) (D) (ii), (iii) and (iv)

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Q54. AIPMT 2006

The orientation of an atomic orbital is governed by

(A) principal quantum number
(B) azimuthal quantum number
(C) spin quantum number
(D) magnetic quantum number.

Open solution

Q55. AIPMT 2006

Given : The mass of electron is 9.11×10^{-31} kg, Planck constant is 6.626×10^{-34} J s, the uncertainty involved in the measurement of velocity within a distance of 0.1 $\mathring{A}$ is

(A) 5.79×10^{5} m s^{-1}
(B) 5.79×10^{6} m s^{-1}
(C) 5.79×10^{7} m s^{-1}
(D) 5.79×10^{8} m s^{-1}

Open solution

Q56. AIPMT 2005

The energy of second Bohr orbit of the hydrogen atom is -328 kJ mol^{-1}; hence the energy of fourth Bohr orbit would be

(A) $-$ 41 kJ mol^{-1}
(B) -82 kJ mol^{-1}
(C) -164 kJ mol^{-1}
(D) -1312 kJ mol^{-1}

Open solution

Q57. AIPMT 2004

The frequency of radiation emitted when the electron falls from n = 4 to n = 1 in hydrogen atom will be (Given ionization energy of H = 2.18×10^{-18} J $atom^{-1}$ and h = 6.625×10^{-34} J s)

(A) 1.54×10^{15} s^{-1}
(B) 1.03×10^{15} s^{-1}
(C) 3.08×10^{15} s^{-1}
(D) 2.00×10^{15} s^{-1}

Open solution

Q58. AIPMT 2003

The value of Planck's constant is 6.63×10^{-34} J s.

The velocity of light is 3.0×10^{8} m s^{-1}. Which value is closest to the wavelength in nanometers of a quantum of light with frequency of 8×10^{15} s^{-1} ?

(A) 2×10^{-25}
(B) 5×10^{-18}
(C) 4×10^{1}
(D) 3×10^{7}

Open solution

Q59. AIPMT 2002

In hydrogen atom, energy of first excited state is $-$ 3.4 eV. Then find out K.E. of same orbit of hydrogen atom

(A) +3.4 eV
(B) +6.8 eV
(C) $-$ 13.6 eV
(D) +13.6 eV

Open solution

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Q60. AIPMT 2001

Main axis of a diatomic molecule is z, molecular orbital p_x and p_y overlap to form which of the following orbitals.

(A) π molecular orbital
(B) σ molecular orbital
(C) δ molecular orbital
(D) No bond will form

Open solution

Q61. AIPMT 2001

The following quantum numbers are possible for how many orbitals :

n = 3, l = 2, m = +2 ?

(A) 1
(B) 2
(C) 3
(D) 4

Open solution

Q62. AIPMT 2000

For given energy, E = 3.03×10^{-19} Joules corresponding wavelength is (h = 6.626×10^{-34} J sec, c = 3×10^8 m/sec)

(A) 65.6 nm
(B) 6.56 nm
(C) 3.4 nm
(D) 656 nm

Open solution

Q63. AIPMT 2000

Isoelectronic species are

(A) CO, CN^-, NO^+, C_2^{2-}
(B) CO^-, CN, NO,
(C) C_2^-
(D) CO, CN, NO, C_2

Open solution

Q64. AIPMT 1999

The uncertainty in momentum of an electron is $1 \times 10^{-5}\ \mathrm{kg\,m/s}$. The uncertainty in its position will be ($h = 6.62 \times 10^{-34}\ \mathrm{kg\,m^2/s}$)

(A) $1.05 \times 10^{-28}\ \mathrm{m}$
(B) $1.05 \times 10^{-26}\ \mathrm{m}$
(C) $5.27 \times 10^{-30}\ \mathrm{m}$
(D) $5.25 \times 10^{-28}\ \mathrm{m}$

Open solution

Q65. AIPMT 1999

Who modified Bohr's theory by introducing elliptical orbits for electron path?

(A) Hund
(B) Thomson
(C) Rutherford
(D) Sommerfeld

Open solution

Q66. AIPMT 1999

The de-Broglie wavelength of a particle with mass $1\ \mathrm{g}$ and velocity $100\ \mathrm{m/s}$ is

(A) $6.63 \times 10^{-33}\ \mathrm{m}$
(B) $6.63 \times 10^{-34}\ \mathrm{m}$
(C) $6.63 \times 10^{-35}\ \mathrm{m}$
(D) $6.65 \times 10^{-36}\ \mathrm{m}$

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Q67. AIPMT 1999

Which of the following configuration is correct for iron?

(A) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^5$
(B) $1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2 3d^5$
(C) $1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^2 3d^7$
(D) $1s^2, 2s^2 2p^6, 3s^2 3p^6, 3d^6, 4s^2$

Open solution

Q68. AIPMT 1998

Which one is not paramagnetic among the following? [Atomic number : Be = 4, Ne = 10, As = 33, Cl = 17]

(A) Cl^- (B) Be^+ (C) Ne^{2+} (D) As^+

Open solution

Q69. AIPMT 1998

Bohr radius for the hydrogen atom ($n = 1$) is approximately 0.530 Å. The radius for the first excited state ($n = 2$) orbit is (in Å)

(A) 0.13 (B) 1.06 (C) 4.77 (D) 2.12

Open solution

Q70. AIPMT 1998

The position of both, an electron and a helium atom is known within 1.0 mm. Further the momentum of the electron is known within 5.0×10^{-26} kg m s^{-1}. The minimum uncertainty in the measurement of the momentum of the helium atom is

(A) 50 kg m s^{-1}
(B) 80 kg m s^{-1}
(C) 80×10^{-26} kg m s^{-1}
(D) 5.0×10^{-26} kg m s^{-1}

Open solution

Q71. AIPMT 1997

The ion that is isoelectronic with CO is

(A) CN^- (B) N_2^+ (C) O^{2-} (D) N_2^-

Open solution

Q72. AIPMT 1996

What will be the longest wavelength line in Balmer series of spectrum?

(A) 546 nm (B) 656 nm (C) 566 nm (D) 556 nm

Open solution

Q73. AIPMT 1996

In a Bohr's model of an atom, when an electron jumps from $n = 1$ to $n = 3$, how much energy will be emitted or absorbed?

(A) 2.389×10^{-12} ergs
(B) 0.239×10^{-10} ergs
(C) 2.15×10^{-11} ergs
(D) 0.1936×10^{-10} ergs

Open solution

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Q74. AIPMT 1995

Uncertainty in position of an electron (mass $= 9.1 \times 10^{-28}$ g) moving with a velocity of 3×10^4 cm/s accurate upto 0.001% will be (Use $h/(4\pi)$ in uncertainty expression where $h = 6.626 \times 10^{-27}$ erg second)

(A) 5.76 cm (B) 7.68 cm (C) 1.93 cm (D) 3.84 cm

Open solution

Q75. AIPMT 1995

The radius of hydrogen atom in the ground state is 0.53 Å. The radius of Li^{2+} ion (atomic number $= 3$) in a similar state is

(A) 0.53 Å (B) 1.06 Å (C) 0.17 Å (D) 0.265 Å

Open solution

Q76. AIPMT 1994

For which of the following sets of four quantum numbers, an electron will have the highest energy?

(A) $n = 3,\ l = 2,\ m = 1,\ s = +1/2$ (B) $n = 4,\ l = 2,\ m = -1,\ s = +1/2$
(C) $n = 4,\ l = 1,\ m = 0,\ s = -1/2$ (D) $n = 5,\ l = 0,\ m = 0,\ s = -1/2$

Open solution

Q77. AIPMT 1994

Which one of the following is not isoelectronic with O^{2-}?

(A) Tl^+ (B) Na^+ (C) N^{3-} (D) F^-

Open solution

Q78. AIPMT 1992

Electronic configuration of calcium atom can be written as

(A) $[Ne]\,4p^2$ (B) $[Ar]\,4s^2$ (C) $[Ne]\,4s^2$ (D) $[Kr]\,4p^2$

Open solution

Q79. AIPMT 1992

The energy of an electron in the nth Bohr orbit of hydrogen atom is

(A) $\frac{13.6}{n^4}$ eV (B) $\frac{13.6}{n^3}$ eV (C) $\frac{13.6}{n^2}$ eV (D) $\frac{13.6}{n}$ eV

Open solution

Q80. AIPMT 1992

If an isotope of hydrogen has two neutrons in its atom, its atomic number and atomic mass number will respectively be

(A) 2 and 1 (B) 3 and 1 (C) 1 and 1 (D) 1 and 3

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Q81. AIPMT 1991

In a given atom no two electrons can have the same values for all the four quantum numbers. This is called

(A) Hund's Rule
(B) Aufbau principle
(C) Uncertainty principle
(D) Pauli's Exclusion principle.

Open solution

Q82. AIPMT 1991

For azimuthal quantum number $l = 3$, the maximum number of electrons will be

(A) 2
(B) 6
(C) 0
(D) 14

Open solution

Q83. AIPMT 1991

The order of filling of electrons in the orbitals of an atom will be

(A) $3d,\ 4s,\ 4p,\ 4d,\ 5s$
(B) $4s,\ 3d,\ 4p,\ 5s,\ 4d$
(C) $5s,\ 4p,\ 3d,\ 4d,\ 5s$
(D) $3d,\ 4p,\ 4s,\ 4d,\ 5s$

Open solution

Q84. AIPMT 1991

The electronic configuration of Cu (atomic number 29) is

(A) $1s^2,\ 2s^2\,2p^6,\ 3s^2\,3p^6,\ 4s^2\,3d^9$
(B) $1s^2,\ 2s^2\,2p^6,\ 3s^2\,3p^6\,3d^{10},\ 4s^1$
(C) $1s^2,\ 2s^2\,2p^6,\ 3s^2\,3p^6,\ 4s^2\,4p^6,\ 5s^2\,5p^1$
(D) $1s^2,\ 2s^2\,2p^6,\ 3s^2\,3p^6,\ 4s^2\,4p^6\,3d^3$

Open solution

Q85. AIPMT 1990

The total number of electrons that can be accommodated in all the orbitals having principal quantum number 2 and azimuthal quantum number 1 are

(A) 2
(B) 4
(C) 6
(D) 8

Open solution

Q86. AIPMT 1990

An ion has 18 electrons in the outermost shell, it is

(A) Cu^+
(B) Th^{4+}
(C) Cs^+
(D) K^+

Open solution

Q87. AIPMT 1989

Which of the following statements do not form a part of Bohr's model of hydrogen atom?

(A) Energy of the electrons in the orbits are quantized.
(B) The electron in the orbit nearest the nucleus has the lowest energy.
(C) Electrons revolve in different orbits around the nucleus.
(D) The position and velocity of the electrons in the orbit cannot be determined simultaneously.

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Q88. AIPMT 1989

Number of unpaired electrons in N^{2+} is/are

(A) 2 (B) 0 (C) 1 (D) 3

Open solution

Q89. AIPMT 1989

The maximum number of electrons in a subshell is given by the expression

(A) $4l - 2$ (B) $4l + 2$ (C) $2l + 2$ (D) $2n^2$

Open solution

Q90. AIPMT 1988

The spectrum of He is expected to be similar to that

(A) H (B) Li^+ (C) Na (D) He^+

Open solution

Q91. AIPMT 1988

The number of spherical nodes in $3p$ orbitals are/is

(A) one (B) three (C) none (D) two

Open solution

Q92. AIPMT 1988

If r is the radius of the first orbit, the radius of nth orbit of H-atom is given by

(A) $r_0 n^2$ (B) $r_0 n$ (C) r_0/n (D) $r_0^2 n^2$

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Answer Key

1. **D**	2. **C**	3. **A**	4. **B**	5. **C**
6. **B**	7. **C**	8. **A**	9. **B**	10. **B**
11. **A**	12. **C**	13. **D**	14. **A**	15. **A**
16. **D**	17. **A**	18. **B**	19. **A**	20. **B**
21. **D**	22. **C**	23. **C**	24. **D**	25. **C**
26. **C**	27. **B**	28. **C**	29. **B**	30. **C**
31. **D**	32. **C**	33. **A**	34. **D**	35. **B**
36. **B**	37. **A**	38. **B**	39. **B**	40. **A**
41. **A**	42. **A**	43. **C**	44. **C**	45. **B**
46. **B**	47. **A**	48. **C**	49. **D**	50. **B**
51. **C**	52. **C**	53. **B**	54. **D**	55. **B**
56. **B**	57. **C**	58. **C**	59. **A**	60. **A**
61. **A**	62. **D**	63. **A**	64. **C**	65. **D**
66. **A**	67. **D**	68. **A**	69. **D**	70. **D**
71. **A**	72. **B**	73. **D**	74. **C**	75. **C**
76. **B**	77. **A**	78. **B**	79. **C**	80. **D**
81. **D**	82. **D**	83. **B**	84. **B**	85. **C**
86. **A**	87. **D**	88. **C**	89. **B**	90. **B**
91. **A**	92. **A**			

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